

projet Holodeck

Déploiement et sécurisation d'une infrastructure réseau Debian à deux machines sans sudo, intégrant un serveur passerelle avec pare-feu nftables, multi-versioning PHP sous Nginx chiffré en HTTPS wildcard, gestion de bases de données, transfert FTPS confiné et authentification centralisée OpenLDAP pour le client.

Configuration réseau et routage du serveur

cat /etc/network/interfaces

```
root@debian:~# cat /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens33
iface ens33 inet dhcp

auto ens37
iface ens37 inet static
    address 192.168.42.1
    netmask 255.255.255.0
root@debian:~#
```

/etc/sysctl.conf

Activer le routage IP

```
#Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
```

Services d'infrastructure : DHCP, DNS et Pare-feu

Installe dnsmasq

configure /etc/dnsmasq.conf

```
interface=ens37
bind-interfaces

dhcp-range=192.168.42.100,192.168.42.200,255.255.255.0,12h

dhcp-option=option:router,192.168.42.1
dhcp-option=option:dns-server,192.168.42.1

domain=starfleet.lan

address=/starfleet.lan/192.168.42.1
address=/www8.starfleet.lan/192.168.42.1
address=/www7.starfleet.lan/192.168.42.1
address=/php.starfleet.lan/192.168.42.1
address=/admin.starfleet.lan/192.168.42.1
root@debian:~#
```

resultat coté client

```
caleb@debian-client:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
state UP group default qlen 1000
    link/ether 00:0c:29:22:25:ca brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.42.147/24 brd 192.168.42.255 scope global dynamic n
oprefixroute ens33
        valid_lft 41385sec preferred_lft 41385sec
    inet6 fe80::20c:29ff:fe22:25ca/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
caleb@debian-client:~$
```

Pare-feu NFTables

```
root@debian:~# cat /etc/nftables.conf
#!/usr/sbin/nft -f
flush ruleset

table inet filter {
    chain input {
        type filter hook input priority 0; policy drop;

        iif "lo" accept
        ct state established,related accept

        iif "ens37" tcp dport { 22, 53, 80, 443, 21, 309, 636, 9090 } accept
        iif "ens37" udp dport { 53, 67 } accept

        iif "ens37" tcp dport 40000-40100 accept
        tcp dport 22 accept
        iif "ens37" ip protocol icmp accept
    }

    chain forward {
        type filter hook forward priority 0; policy drop;

        iif "ens37" oif "ens33" accept
        iif "ens33" oif "ens37" ct state established,related accept
    }
}

table ip nat {
    chain postrouting {
        type nat hook postrouting priority 100; policy accept;

        oif "ens33" masquerade
    }
}

root@debian:~#
```

Pare-feu NfTables

```
root@debian:~# cat /etc/nftables.conf
#!/usr/sbin/nft -f
flush ruleset

table inet filter {
    chain input {
        type filter hook input priority 0; policy drop;

        ilf "lo" accept
        ct state established,related accept

        ilf "ens37" tcp dport { 22, 53, 80, 443, 21, 389, 636, 9090 } accept
        ilf "ens37" udp dport { 53, 67 } accept

        ilf "ens37" tcp dport 40000-40100 accept

        tcp dport 22 accept

        ilf "ens37" ip protocol icmp accept
    }

    chain forward {
        type filter hook forward priority 0; policy drop;

        ilf "ens37" oif "ens33" accept
        ilf "ens33" oif "ens37" ct state established,related accept
    }
}

table ip nat {
    chain postrouting {
        type nat hook postrouting priority 100; policy accept;

        oif "ens33" masquerade
    }
}
root@debian:~#
```

Nginx, MariaDB, PHP 7.4 et 8.3

wget https://exemple.com

```
root@debian:~# nginx -v
mariadb --version
php7.4 -v
php8.3 -v
nginx version: nginx/1.31.4
mariadb from 12.3.3-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using EditLine wrapper
PHP 7.4.33 (cli) (built: Jul 3 2026 10:36:02) ( NTS )
Copyright (c) The PHP Group
Zend Engine v3.4.0, Copyright (c) Zend Technologies
    with Zend OPcache v7.4.33, Copyright (c), by Zend Technologies
PHP 8.3.33 (cli) (built: Jul 30 2026 15:43:59) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.33, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.33, Copyright (c), by Zend Technologies
root@debian:~#
```

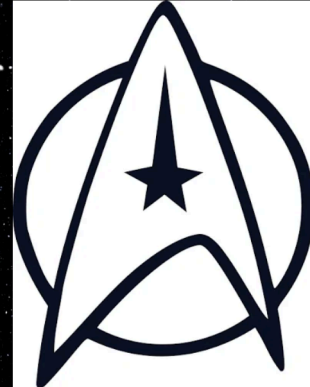
Certificat de sécurité SSL

creation des fichiers

```
mkdir -p /etc/ssl/starfleet  
chmod 700 /etc/ssl/starfleet  
chmod 600 /etc/ssl/starfleet/starfleet.key  
chmod 644 /etc/ssl/starfleet/starfleet.crt
```

générer le certificat SSL wildcard

```
openssl req -x509 -nodes -days 365 -newkey rsa:2048 \  
-keyout /etc/ssl/starfleet/starfleet.key \  
-out /etc/ssl/starfleet/starfleet.crt \  
-subj "/CN=*.starfleet.lan" \  
-addext "subjectAltName=DNS:*.starfleet.lan,DNS:starfleet.lan"
```



Création de l'arborescence web

```
mkdir -p /var/www/starfleet/www7  
mkdir -p /var/www/starfleet/www8
```

```
cat /var/www/starfleet/www7/index.php
```

```
root@debian:~# cat /var/www/starfleet/www7/index.php  
<?php  
echo "<h1>Bienvenue sur Starfleet - PHP 7</h1>";  
echo "<p>Version executée : " . phpversion() . "</p>";
```

```
chown -R www-data:www-data /var/www/starfleet  
chmod -R 755 /var/www/starfleet
```

Configurer Nginx pour www7 et www8

nano /etc/nginx/conf.d/www7.conf

```
GNU nano 7.2 /etc/nginx/conf.d/www7.conf
server {
    listen 80;
    server_name www7.starfleet.lan;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name www7.starfleet.lan;

    ssl_certificate /etc/ssl/starfleet/starfleet.crt;
    ssl_certificate_key /etc/ssl/starfleet/starfleet.key;

    root /var/www/starfleet/www7;
    index index.php index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        fastcgi_pass unix:/run/php/php7.4-fpm.sock;
    }
}
```

Bienvenue sur Starfleet - PHP 7
Version executée : 7.4.33

nano /etc/nginx/conf.d/www8.conf

```
GNU nano 7.2 /etc/nginx/conf.d/www8.conf
server {
    listen 80;
    server_name www8.starfleet.lan;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name www8.starfleet.lan;

    ssl_certificate /etc/ssl/starfleet/starfleet.crt;
    ssl_certificate_key /etc/ssl/starfleet/starfleet.key;

    root /var/www/starfleet/www8;
    index index.php index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
        fastcgi_param HTTP_AUTHORIZATION $http_authorization;
    }
}
```

Bienvenue sur Starfleet - PHP 8
Version executée : 8.3.33

phpmyadmin

```
mkdir -p /var/www/starfleet/phpmyadmin/tmp
chown -R www-data:www-data /var/www/starfleet/phpmyadmin
chmod 770 /var/www/starfleet/phpmyadmin/tmp
```

```
GNU nano 7.2 /var/www/starfleet/phpmyadmin/config.inc.php
<?php
/*
 * phpmyadmin sample configuration, you can use it as base for
 * manual configuration. For easier setup you can use setup!
 *
 * All directives are explained in documentation in the doc/ folder
 * or at <https://docs.phpmyadmin.net/>.
 */

declare(strict_types=1);

/*
 * This is needed for cookie based authentication to encrypt the cookie.
 * Needs to be a 32-bytes long string of random bytes. See FAQ 2.10.
 */
$config['blowfish_secret'] = 'StarfleetSecretKeyEnterprise00261'; /* YOU MUST FILL IN THIS FOR COOKIE AUTH! */

/*
 * Servers configuration
 */
$i = 0;

/*
 * First server
 */
$i++;
/* Authentication type */
$config['Servers'][$i]['auth_type'] = 'cookie';
/* Server parameters */
$config['Servers'][$i]['host'] = 'localhost';
$config['Servers'][$i]['compress'] = false;
$config['Servers'][$i]['AllowUnsafeLinks'] = false;

/*
 * phpmyadmin configuration storage settings.
 */
/* User used to manipulate with storage */
// $config['Servers'][$i]['controlhost'] = '';
// $config['Servers'][$i]['controluser'] = 'pma';
// $config['Servers'][$i]['controlpass'] = 'pma';
// $config['Servers'][$i]['controlsock'] = 'pma';
/* Storage database and tables */
// $config['Servers'][$i]['table'] = 'phpmyadmin';
// $config['Servers'][$i]['bookmarktable'] = 'pma_bookmark';
```

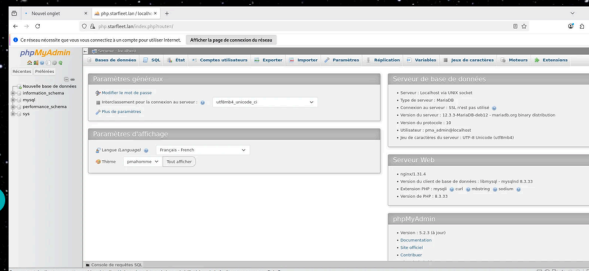
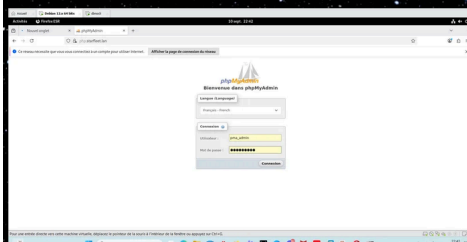
l'écriture de l'ID de l'utilisateur

créer le fichier `/etc/nginx/conf.d/phpmyadmin.conf`

```
GNU nano 2.2 /etc/nginx/conf.d/phpmyadmin.conf
server {
    listen 80;
    server_name php.starfleet.lan;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name php.starfleet.lan;
    ssl_certificate /etc/ssl/starfleet/starfleet.crt;
    ssl_certificate_key /etc/ssl/starfleet/starfleet.key;
    root /var/www/starfleet/phpmyadmin;
    index index.php index.html;
    client_max_body_size 60M;
    location / {
        try_files $uri /index.html;
    }

    location ~ /\.php$ {
        include fastcgi_params;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        fastcgi_pass unix:/run/php/php7.2-fpm.sock;
    }
}
```



Cockpit

Active et démarre le service
`systemctl enable --now cockpit.socket`

il faut lui indiquer de faire confiance à `admin.starfleet.lan`.

`nano /etc/cockpit/cockpit.conf`

```
GNU nano 7.2 /etc/cockpit/cockpit.conf
[WebService]
Origins = https://admin.starfleet.lan
ProtocolHeader = X-Forwarded-Proto
AllowUnencrypted = true
```

`systemctl restart cockpit`

Créer le Virtual Host Nginx (admin.starfleet.lan)

les relayer les connexion nginx de manière transparente vers le port interne de Cockpit (9090)

nano /etc/nginx/conf.d/admin.conf

```
root@debian:~# cat /etc/nginx/conf.d/admin.conf
server {
    listen 80;
    server_name admin.starfleet.lan;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name admin.starfleet.lan;

    ssl_certificate /etc/ssl/starfleet/starfleet.crt;
    ssl_certificate_key /etc/ssl/starfleet/starfleet.key;
    ssl_protocols TLSv1.2 TLSv1.3;

    location / {
        proxy_pass http://127.0.0.1:9090;

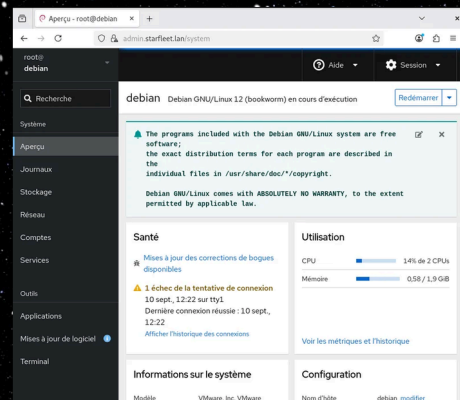
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";

        proxy_read_timeout 86400;
    }
}

root@debian:~#
```

systemctl restart nginx



vsftpd

Créer l'utilisateur FTP dédié

```
useradd -m -d /var/www/starfleet -s /bin/bash devweb
passwd devweb
```

Ajoutons ensuite devweb au groupe www-data pour qu'il puisse éditer les fichiers web sans conflit de droits

```
usermod -aG www-data devweb
```

Configurer vsftpd

nano /etc/vsftpd.conf

```
GNU nano 2.2 /etc/vsftpd.conf
listen=YES
listen_ipv6=NO
anonymous_enable=NO
local_enable=YES
#ft_enable=YES
local_umask=022
chroot_local_user=YES
allow_writeable_chroot=YES
#ssv_enable=YES
#ssv_max_port=4000
#ssv_max_port=40100
ssl_enable=YES
#ssl_min_version=SSLv3
force_local_data_ssl=YES
force_local_logins_ssl=YES
#ssl_cert=/etc/ssl/starfleet/starfleet.crt
#ssl_private_key=/etc/ssl/starfleet/starfleet.key
ssl_tlsv1=YES
ssl_sslv2=NO
ssl_sslv3=NO
require_ssl_reuse=NO
ssl_ciphers=ALL
```

```
caleb@debian-client:~$ lftp -u devweb starfleet.lan
Mot de passe :
```

```
lftp devweb@starfleet.lan:~> ls
-rw-r--r--  1 0      0      1104 Sep 10 10:35 auth_ldap.ph
p
drwxr-xr-x 13 33     33     4096 Sep 11 07:16 phpmyadmin
drwxr-xr-x  2 33     33     4096 Sep 07 07:02 www7
drwxr-xr-x  2 33     33     4096 Sep 10 10:43 www8
lftp devweb@starfleet.lan:~>
```

déployer l'annuaire OpenLDAP

dpkg-reconfigure slapd

mettre la configuration d'OpenLDAP ? → Non
Nom de domaine DNS : tapez starfleet.lan
Nom de l'entité / organisation : tapez Starfleet
Mot de passe administrateur : retapez votre mot de passe
Confirmer le mot de passe : retapez-le à nouveau
Faut-il supprimer la base lorsque slapd est purgé ? → Non
Déplacer l'ancienne base de données ? → Oui

nano /tmp/structure.ldif

```
nano /tmp/structure.ldif
```

```
dn: ou=users,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: users

dn: ou=groups,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: groups
```

la fiche d'identité complète de l'utilisateur dont l'identifiant est caleb.

```
root@debian:~# ldapsearch -x -b "dc=starfleet,dc=lan" "(uid=caleb)"
# extended LDIF
#
# LDAPv3
# base <dc=starfleet,dc=lan> with scope subtree
# filter: (uid=caleb)
# requesting: ALL
#
# caleb, users, starfleet.lan
dn: uid=caleb,ou=users,dc=starfleet,dc=lan
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: caleb
sn: olago
givenName: Caleb
cn: Caleb olago
displayName: Caleb olago
uidNumber: 10001
gidNumber: 10001
loginShell: /bin/bash
homeDirectory: /home/caleb

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
root@debian:~#
```

```
ldapadd -x -D "cn=admin,dc=starfleet,dc=lan" -W -f /tmp/structure.ldif
```

les pages de connexion LDAP sur www8 et www7

Nous allons remplacer la page d'accueil
de www8.starfleet.lan par un script

nano /var/www/starfleet/www8/index.php

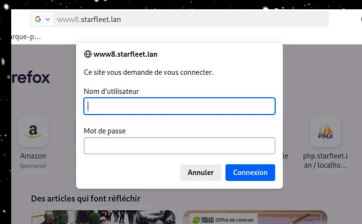
```
root@debian:~# cat /var/www/starfleet/www8/index.php<?php
if (isset($_SERVER['PHP_AUTH_USER'])) {
    header('WWW-Authenticate: Basic realm="Acces Restreint Starfleet"');
    header('HTTP/1.0 401 Unauthorized');
    echo "Authentication requise";
    exit;
}

$user = $_SERVER['PHP_AUTH_USER'];
$password = $_SERVER['PHP_AUTH_PW'];

$ldap_conn = ldap_connect("localhost", 389);
ldap_set_option($ldap_conn, LDAP_OPT_PROTOCOL_VERSION, 3);
$user_dn = "uid=" . $user . ",ou=users,dc=starfleet,dc=lan";
$auth = @ldap_bind($ldap_conn, $user_dn, $password);

if ($auth) {
    echo "<div>Connexion LDAP reussie pour " . htmlspecialchars($user) . " </div>";
} else {
    header('WWW-Authenticate: Basic realm="Acces Restreint Starfleet"');
    header('HTTP/1.0 401 Unauthorized');
    echo "Identifiants invalides";
}
ldap_close($ldap_conn);
?>
root@debian:~#
```

connexion chez le client





fin !

